

AI in Real World Context

A 60-minute field guide to tools, services and next-token predictors

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1. Prologue: AI in the Real World

AI in Real → Current World Context

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- ◇ **Real** promises a fixed world — it isn't one.

AI in Real → Current World Context

- ◇ **Real** promises a fixed world — it isn't one.
- ◇ **Current**: what exists *today*, in your browser, your phone, your lab.

AI in Real → Current World Context

- ◇ **Real** promises a fixed world — it isn't one.
- ◇ **Current**: what exists *today*, in your browser, your phone, your lab.
- ◇ Everything in this talk is something you can touch **this week**.

so, let's start with you

What do you mean by “AI”?

so, let's start with you

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◇ ChatGPT / Gemini / Copilot?

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What do you mean by “AI”?

- ◇ ChatGPT / Gemini / Copilot?
- ◇ Robots & self-driving cars?

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- ◇ The algorithm that recommends your reels?

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- ◇ A calculator on steroids?

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- ◇ Pure hype?

so, let's start with you

What do you mean by “AI”?

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- ◇ Pure hype?

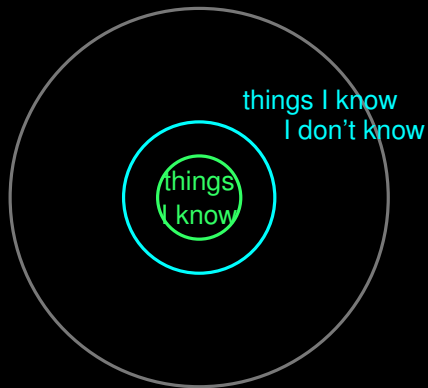
the deal

Whatever you said — we'll build on it; by the end you'll have a sharper answer.

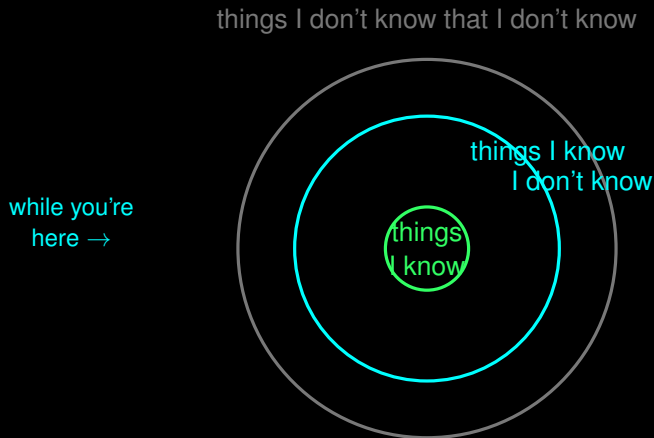
2. The Agenda

The room, mapped | three circles of knowing

things I don't know that I don't know



The room, mapped | three circles of knowing



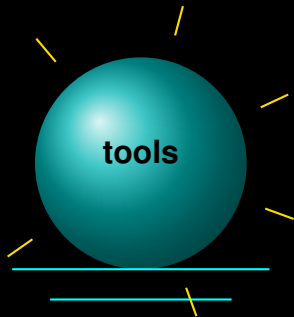
Objective: increase the area of **things you know you don't know**.

The room, mapped | three circles of knowing

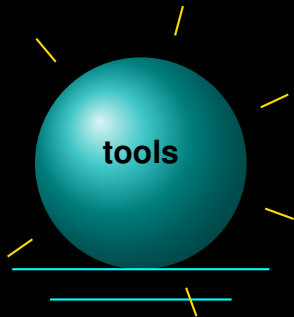


Objective: increase the area of **things you know you don't know**.

I had a premonition | and it was about tools

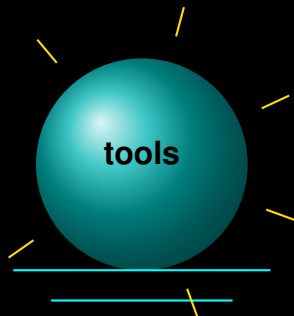


I had a premonition | and it was about tools



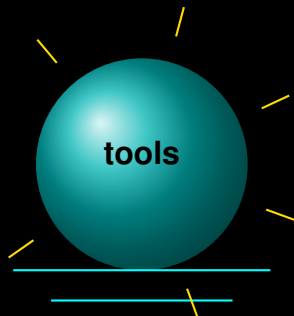
- ◇ The audience wants **tools**, not theory.

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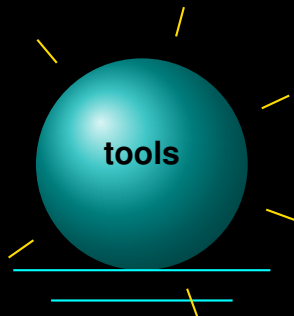
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- ◇ I got the premonition.

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- ◇ And the agenda wrote itself:

I had a premonition | and it was about tools



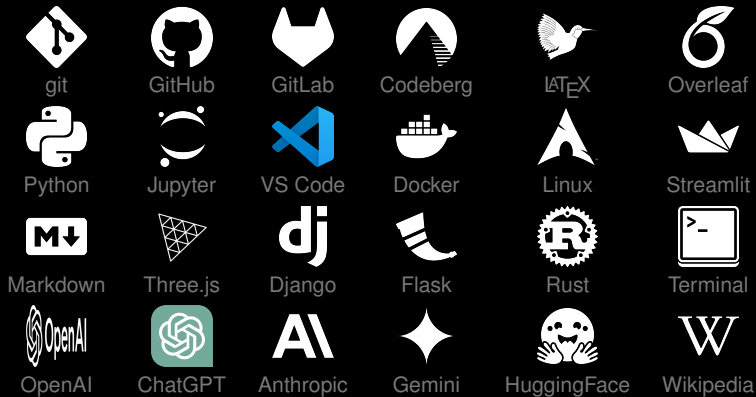
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Goal #1

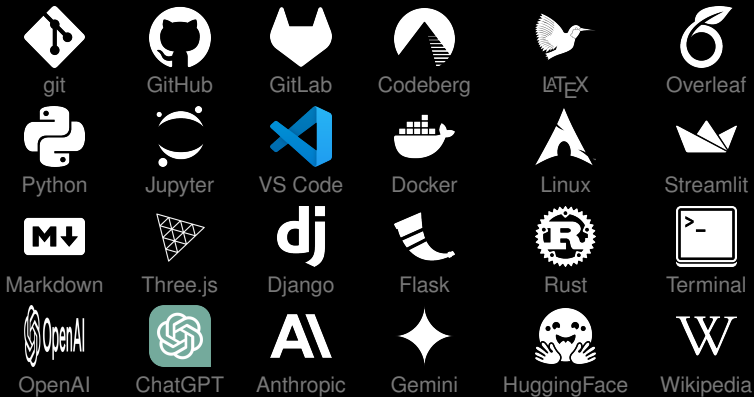
In the next **1 hour**, you will learn about tools.

3. Tool Time

What kind of tools? | spoiler: not all of these are AI



What kind of tools? | spoiler: not all of these are AI



Are all these AI tools?

Most of them aren't AI.

Git, $\text{\LaTeX}$, the terminal — older than most of you, zero AI.
The AI ones (ChatGPT, Gemini, Claude) *sit on top* of these.

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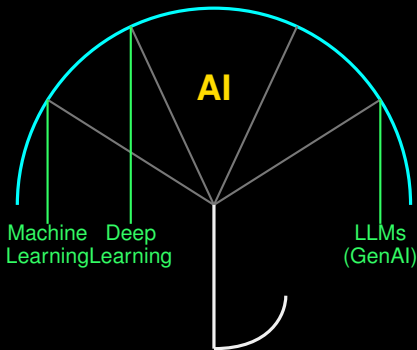
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the point

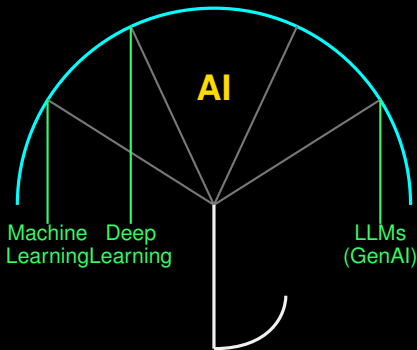
The non-AI tools are **leverage**: learn them once, use them forever.
With them, the AI tools become **superpowers** instead of toys.

4. The AI Landscape

AI is an umbrella term | what I work in: LLMs



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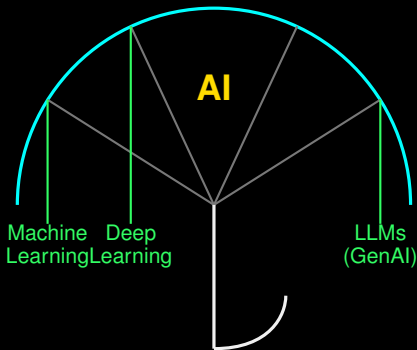


- ◇ **AI** = the umbrella over everything "intelligent".



xkcd 1838 — machine learning, the honest version

AI is an umbrella term | what I work in: LLMs

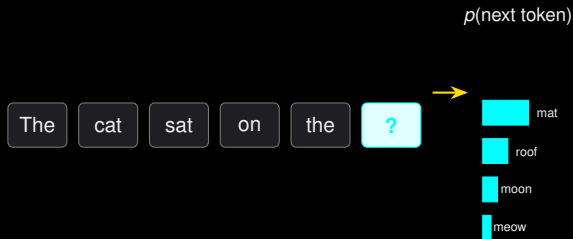


- ◇ **AI** = the umbrella over everything "intelligent".
- ◇ What I work in: **LLMs** — *next-token predictors*.



xkcd 1838 — machine learning, the honest version

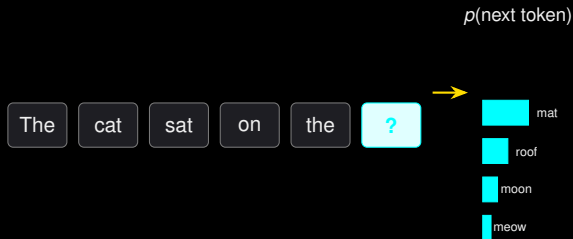
LLMs = next-token predictors | everything ChatGPT does, one loop



Input: a token sequence $\longrightarrow$ **Output:** probabilities over the next token.

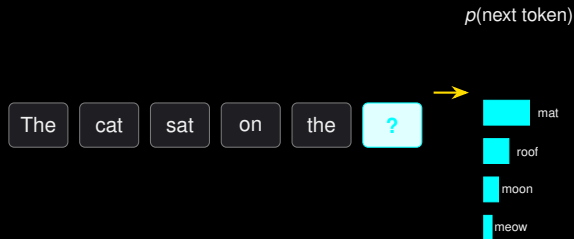
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- ◇ The model never *searches* — it *predicts*.



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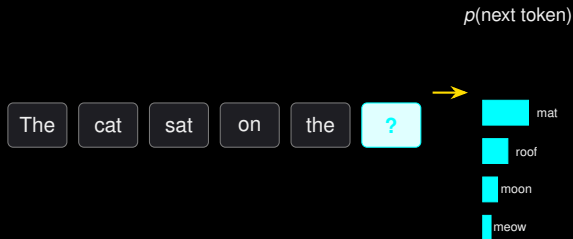
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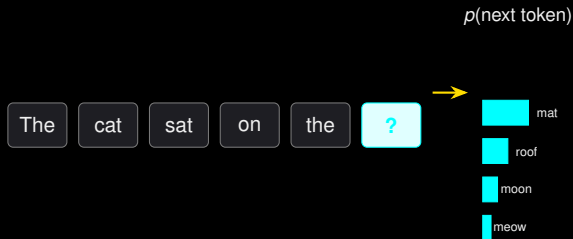
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- ◇ Add tools, and prediction becomes **action**.

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trust me bro — I know this token

Through the lens | once you see it, you can't unsee it

deep learning AGI learning
chatbots agents pe

**next token
prediction**

$$p(w_t \mid w_{<t})$$

Through the lens | once you see it, you can't unsee it

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**next token
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$$p(w_t | w_{<t})$$

- ◇ The **lens**: look at any AI through it.

Through the lens | once you see it, you can't unsee it

deep learning AGI reasoning
chatbots agents type

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- ◇ The **lens**: look at any AI through it.
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- ◇ The **lens**: look at any AI through it.
- ◇ **Agents**? next-token prediction + tools.
- ◇ **Alignment**? steering the same predictor.

Through the lens | once you see it, you can't unsee it

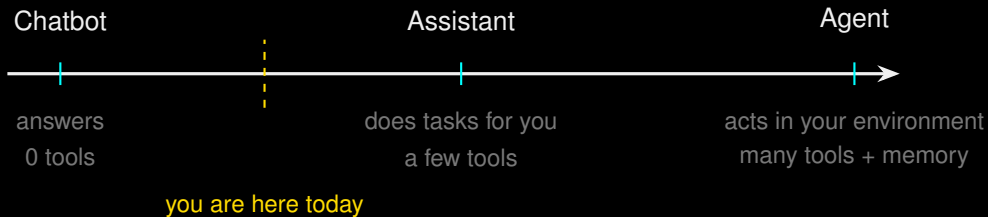
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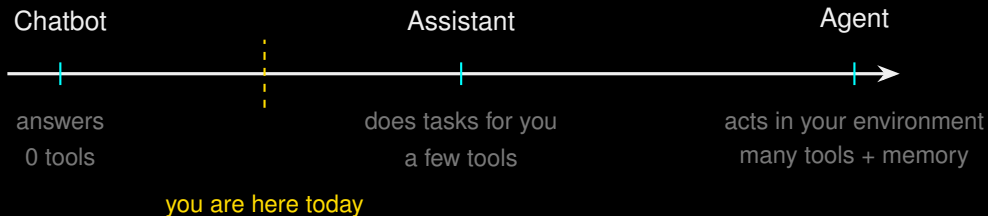
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- ◇ The **lens**: look at any AI through it.
- ◇ **Agents**? next-token prediction + tools.
- ◇ **Alignment**? steering the same predictor.
- ◇ The hype is loud; the mechanism is simple.

The assistant axis | from chatbot to agent



The assistant axis | from chatbot to agent



Question: where do you want to live on this axis by the end of the year?

The alignment wall | a tiny corner of a huge research effort

Real research is happening on **AI alignment** — nine papers I keep going back to:

Concrete Problems in AI Safety

Amodei, Olah, et al.

2016 · [arXiv:1606.06565](#)

AI Safety via Debate

Irving, Christiano, Amodei

2018 · [arXiv:1805.00899](#)

Scalable Agent Alignment via Reward Modeling

Leike, Krueger, et al.

2018 · [arXiv:1811.07871](#)

Fine-Tuning LMs from Human Preferences

Ziegler, Stiennon, et al.

2019 · [arXiv:1909.08593](#)

Training a Helpful & Harmless Assistant (RLHF)

Bai, Jones, et al.

2022 · [arXiv:2204.05862](#)

Constitutional AI: Harmlessness from AI Feedback

Bai, Kadavath, et al.

2022 · [arXiv:2212.08073](#)

Towards Understanding Syco- phancy

Sharma, Tong, et al.

2023 · [arXiv:2310.13548](#)

Weak-to-Strong Generalization

Burns, Izmailov, et al.

2023 · [arXiv:2312.09390](#)

The AI Alignment Paradox

West & Aydin

2024 · [arXiv:2405.20806](#)

every paper is free on arXiv — pick one, read it tonight

The paper that started this fourth wave | Attention Is All You Need



Computer Science > Computation and Language

[Submitted on 12 Jun 2017 (v1), last revised 2 Aug 2023 (this version, v7)]

Attention Is All You Need

[Ashish Vaswani](#), [Noam Shazeer](#), [Niki Parmar](#), [Jakob Uszkoreit](#), [Llion Jones](#), [Aidan N. Gomez](#), [Lukasz Kaiser](#), [Illia Polosukhin](#)

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Comments: 15 pages, 5 figures

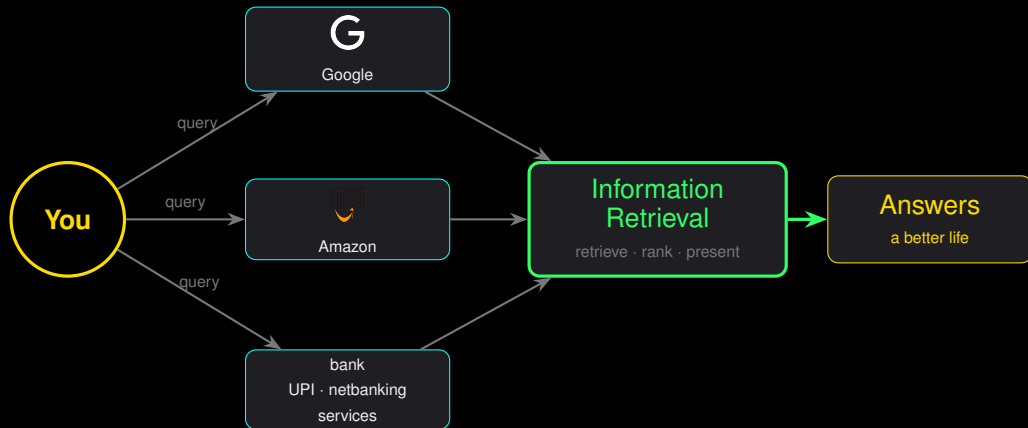
Subjects: **Computation and Language (cs.CL)**; Machine Learning (cs.LG)

Cite as: [arXiv:1706.03762](#) [cs.CL]
(or [arXiv:1706.03762v7](#) [cs.CL] for this version)
<https://doi.org/10.48550/arXiv.1706.03762> 

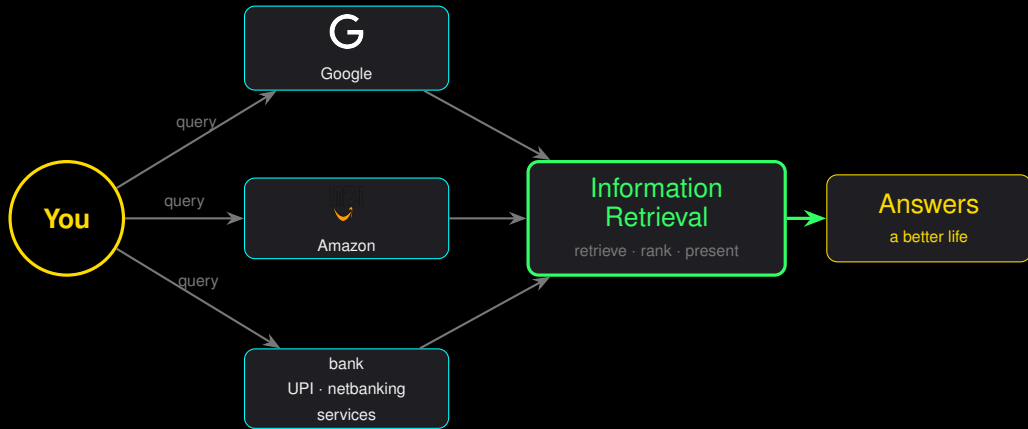
Vaswani et al., 2017 — **Attention Is All You Need**
arXiv:1706.03762

5. Fundamentals

How do you interact with computers? | spoiler: services



How do you interact with computers? | spoiler: services



Google, Amazon, even banks — they are **services**.
What do services do? **They retrieve information for you** — and make your life better.

Shuvam Banerji Seal

BS–MS final-year student, IISER Kolkata
Chemistry major · Computer Science minor
Research: **Information Retrieval** — RAG,
BM25, TREC, ECIR'26
Clubs: President, Slashdot · Office Bearer,
Valence

github.com/Shuvam-Banerji-Seal · shuvam-banerji-seal.github.io

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- ◇ I study chemistry, I code for a living, I research **retrieval**.
- ◇ I teach tools for fun — like this talk.
- ◇ Everything I show you, I use daily.

You tell me.

You tell me.

?



Video

most of the bytes



Photos

many bytes



Text

few bytes, most value

You tell me.

?



Video

most of the bytes



Photos

many bytes

W

Text

few bytes, most value

out of the three — **text** is the one that constitutes all these services and softwares.

It's all text | the universal interface

- ◇ Out of **video**, **photos** and **text** — which one runs every service?

It's all text | the universal interface

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It's all text | the universal interface

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- ◇ LLMs eat text. **Agents act on text.**

It's all text | the universal interface

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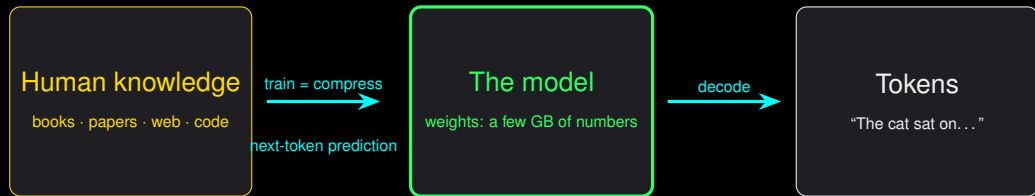
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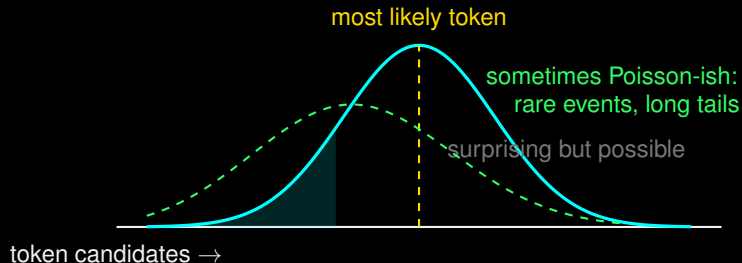
6. Knowledge Compression

LLMs = human knowledge, compressed into a model | the library in a file



A whole library, compressed into one file — that predicts the next word.

The distribution of the next token | a Gaussian (or maybe a Poisson)



The model is a **probability distribution** over the next token — your knowledge, your language, your style, compressed.

Too much maths? | okay, that was the last formula



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- ◇ Don't worry — that was the only formula in this talk.

Too much maths? | okay, that was the last formula



- ◇ Don't worry — that was the only formula in this talk.
- ◇ The takeaway is one line: **LLMs predict the next token.**

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- ◇ Don't worry — that was the only formula in this talk.
- ◇ The takeaway is one line: **LLMs predict the next token.**
- ◇ And now that you know text matters. . .

Too much maths? | okay, that was the last formula



- ◇ Don't worry — that was the only formula in this talk.
- ◇ The takeaway is one line: **LLMs predict the next token.**
- ◇ And now that you know text matters. . .
- ◇ **let's get back to the tools.**

7. Tools You Should Learn

How many of you have a personal website? | hands up!

Hands up!

How many of you have a personal website? | hands up!

Hands up!

Want one? Host it free on:



GitHub



GitLab



Codeberg

Underlying tool: **git**. You need to learn git.

Hands up!

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Codeberg

Underlying tool: **git**. You need to learn git.

\$ your first repo

```
$ git init
$ git add .
$ git commit -m "my first website"
$ git push origin main
```

How many of you have a personal website? | hands up!

Hands up!

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GitHub



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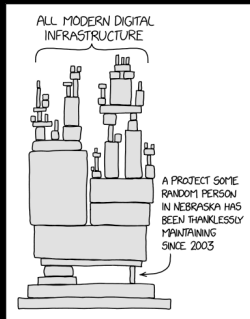


Codeberg

Underlying tool: **git**. You need to learn git.

\$ your first repo

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$ git init
$ git add .
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$ git push origin main
```



xkcd 2347: Dependency — everyone runs on git, eventually

GitHub, in action | the project overview

Shuvam-Banerji-Seal / ai-in-real-world-context

Q

Type (7) to search

Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

ai-in-real-world-context

Public

Pin

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Q

Go to file

T

Add file

Code

Shuvam-Banerji-Seal

HTML: move xkcd Dependency meme from git slide to the O...

7b95f76 · 9 minutes ago

8 Commits

assets

Add 'Attention Is All You Need' slide (HTML+LaTeX); growl...

17 minutes ago

sections

Add 'Attention Is All You Need' slide (HTML+LaTeX); growl...

17 minutes ago

.gitignore

Fix step reveal (hide points until \pause-step), fresh mem...

3 hours ago

.olignore

AI in Real World Context: HTML deck + LaTeX beamer + as...

3 hours ago

LICENSE

AI in Real World Context: HTML deck + LaTeX beamer + as...

3 hours ago

README.md

Fix step reveal (hide points until \pause-step), fresh mem...

3 hours ago

index.html

HTML: move xkcd Dependency meme from git slide to th...

9 minutes ago

log

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main.pdf

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main.tex

AI in Real World Context: HTML deck + LaTeX beamer + as...

3 hours ago

sbs_dark_beamer.sty

AI in Real World Context: HTML deck + LaTeX beamer + as...

3 hours ago

About

AI in Real World Context — a 60-min pointer-friendly HTML deck + LaTeX Beamer (tools, services, next-token predictors).

shuvam-banerji-seal.github.io/ai-l...

Readme

MIT license

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Deployments 8

github-pages 9 minutes ago



Overleaf



L^AT_EX

- ◇ PDFs: assignments, thesis, projects — all L^AT_EX.



Overleaf



L^AT_EX



Overleaf



L^AT_EX

- ◇ PDFs: assignments, thesis, projects — all L^AT_EX.
- ◇ You type text + commands; the machine typesets beauty.



Overleaf



L^AT_EX

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Overleaf



$\text{\LaTeX}$

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fun fact

Heck — even this presentation is made using $\text{\LaTeX}$.
You are literally reading typeset tokens.

Overleaf, in action | the project view

The screenshot displays the Overleaf web editor interface for a LaTeX project titled "Machine Verified Proving of Biological Information". The interface is divided into three main sections:

- Left Panel:** Contains the "File tree" and "File outline". The "File tree" shows a directory structure with files like `01_prologue.tex`, `02_agenda.tex`, `03_tooltime.tex`, `04_aillandscape.tex`, `05_fundamentals.tex`, `06_knowledge.tex` (highlighted), `07_tools.tex`, and `08_agents.tex`. The "File outline" shows a hierarchy of sections, with "Too much maths?" highlighted under the "Knowledge Compression" section.
- Center Panel:** Displays the LaTeX source code for `06_knowledge.tex`. The code includes comments and LaTeX commands such as `\begin{frame}`, `\end{frame}`, `\begin{columns}`, `\begin{column}`, `\includegraphics`, and `\item`. A line of code `\includegraphics[height=4.7cm]{assets/img/one_does_not_simplify.jpg}` is highlighted.
- Right Panel:** Shows the compiled PDF output. The title bar indicates "Overleaf & L^AT_EX | assignments, thesis, projects". The PDF content shows a page with the number 66 / 82 and a 99% zoom level. The main content area displays the rendered text, including the "Too much maths?" section, which features a large number 6 and a bird icon.

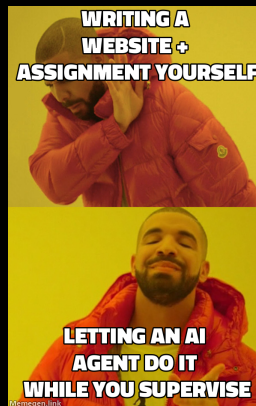
Would you do it by yourself? | now that everything is text

So now I've asked you to build a website and write an assignment.

Would you do it by yourself? | now that everything is text

So now I've asked you to build a website and write an assignment.

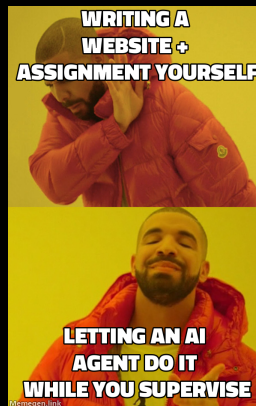
- ◇ Would you want to do it all **by yourself**?



Would you do it by yourself? | now that everything is text

So now I've asked you to build a website and write an assignment.

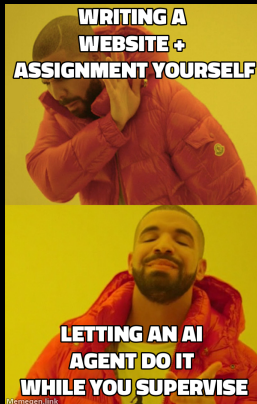
- ◇ Would you want to do it all **by yourself**?
- ◇ Remember: everything here is **text**.



Would you do it by yourself? | now that everything is text

So now I've asked you to build a website and write an assignment.

- ◇ Would you want to do it all **by yourself**?
- ◇ Remember: everything here is **text**.
- ◇ And text is exactly what AI agents are good at.



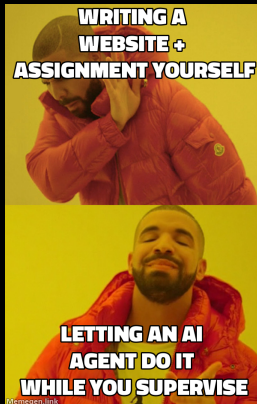
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- ◇ Would you want to do it all **by yourself**?
- ◇ Remember: everything here is **text**.
- ◇ And text is exactly what AI agents are good at.

the pitch

Use **agentic tools** — like the ones I'm about to show.



8. Agentic Tools

Use agentic tools | AI that acts, not just answers

- ◇ Tools I use daily: **VS Code** + **agentic coding tools**.

a session, roughly

```
$ opencode
» write a python script that
» downloads all my course PDFs
✓ plan:  fetch, parse, save
✓ wrote downloads.py
✓ ran it:  42 files saved
$
```

Use agentic tools | AI that acts, not just answers

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- ◇ Open-source, runnable locally: OpenCode, Claude Code, ...

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not sci-fi — this is a Tuesday.

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OpenAI



ChatGPT



Claude



Gemini



HF



shell

a session, roughly

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$
```

not sci-fi — this is a Tuesday.

OpenCode, in action | the project view

```
50 + \begin{center}
51 + \includegraphics[height=6.2cm]{assets/img/
github_project_overview_screenshot.png}
52 + \end{center}
53 + \end{frame}
54 +
46 \subsection{\texorpdfstring{\LaTeX{} \& Overleaf}{LaTeX and
Overleaf}}
47
48 \begin{frame}[Overleaf \& \LaTeX]{assignments, thesis, projects}
49 \begin{columns}
50 \begin{frame}[Overleaf \& \LaTeX]{assignments, thesis, projects}
51 \begin{columns}

Recompiling the LaTeX and verifying zero warnings:

$ cd /home/shuvam/codes/presentation_cbse && for i in 1 2 3; do timeout 240 pdflatex -shell-escape -interaction=nonstopmode main.tex > /tmp/
opencode/z31.log 2>&1; done; echo "errors: $(grep -cE 'A' /tmp/opencode/z31.log)"; echo "Overfull: $(grep -c Overfull /tmp/opencode/z31.log)";
echo "Underfull: $(grep -c Underfull /tmp/opencode/z31.log)"; grep "Warning" /tmp/opencode/z31.log | sort -u | head; pdftotext main.pdf | grep
Pages; echo "=== new slides in PDF? ==="; pdftotext main.pdf - 2>/dev/null | grep -c "GitHub, in action\|Overleaf, in action"

errors: 0
Overfull: 0
Underfull: 0
Pages: 84
=== new slides in PDF? ===
2

Thought: 957ms

LaTeX compiles clean: 0 errors, 0 overfull, 0 underfull, 84 pages. Both new slides (GitHub in action, Overleaf in action) are in the PDF.

Now let me verify the HTML structure (29 sections), then commit everything, push to GitHub, verify Pages, sync Overleaf. Ensure main.pdf is on
the GitHub repo.

Let me do the verification + commit + push + pages + overleaf.

Clean compile - 84 pages, both new slides present. Verifying HTML and shipping everything:

- Writing command...

x Build - DeepSeek V4 Flash (2x usage)

In the html move the  image from the []

Build - DeepSeek V4 Flash (2x usage) OpenCode Go

***** esc interrupt
```

528.0K (53%) · \$3.72 ctrl+p commands

Beamer presentation on AI in real world context

Context

528,041 tokens
53% used
\$3.72 spent

MCP

- arxiv Connected
- context7 Connected
- diagrams-mcp Connected
- drawio Connected
- duckduckgo-search Connected
- everything Connected
- filesystem Connected
- git MCP error -32000: Connection closed
- hyprland Connected
- math-svg Disabled
- mcp-diagram-generator Connected
- memory Connected
- plantuml Connected
- playwright Operation timed out after 3000ms
- sequential-thinking Connected
- shell Connected
- sqlite Connected
- time Connected
- upmath Connected
- wikipedia Connected

LSP

LSPs are disabled

Todo

- [✓] Explore environment (latex tools, assets, converters)
- [✓] Research user (Shuvam, IISER Kolkata) via web search + CV
- [✓] Collect real AI alignment research papers (arXiv)
- [✓] Download logos/images/memes (26 logos, 8 memes, 4 xkcd)
- [✓] Write beamer presentation with

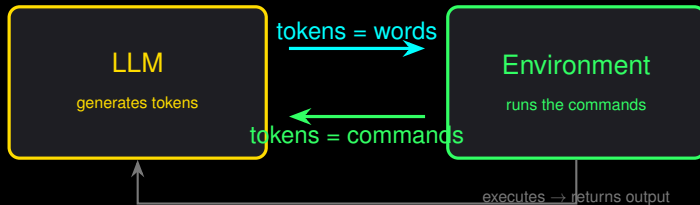
~/codes/presentation_cbse

• OpenCode 1.18.16

What are agentic tools? | tokens become commands



What are agentic tools? | tokens become commands



LLMs generate tokens (words). What if the token is a **command** — and the environment it runs in **executes the command** as well? That's an **agentic tool**.

Questions? Everything is text.

Agents turn text into action.

github.com/Shuvam-Banerji-Seal · shuvam-banerji-seal.github.io
